Marwan Wael Elhusseiny Abdelaziz

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EDUCATION

Sahara International School — *Cairo, Egypt*
IGCSE Student
October 2019 – June 2022

Helwan University — Cairo, Egypt

- Bachelor Degree in Information and Communication Engineering: Computer Engineering
- Expected Graduation: 2027
- GPA: 2.41
- Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Operating Systems, Digital System Design (VHDL, FPGA), Web Development (HTML, CSS), Software Engineering, Programming Paradigms (C++)

SKILLS

Soft Skills: Fast Learner, Critical Thinking, Problem Solving, Adaptability, Team Collaboration, Communication Skills, Emotional Intelligence

Hard Skills: C++ (OOP, Data Structures), Web Development (HTML, CSS), VHDL (FPGA-Based Design)

PROJECTS

Home Security System | ATmega32, C, USART, Proteus
Team Member *Apr 2025 – May 2025*

- Developed a basic home security system using two ATmega32 microcontrollers.
- Implemented intrusion detection using PIR and magnetic sensors, with system states controlled via push button.
- Coded logic for buzzer alarms, LED status indicators, and USART-based communication between microcontrollers.
- Simulated the full setup in Proteus, including real-time monitoring and remote control functionality.

Digital Counter (0–9) | VHDL, Quartus, FPGA
Team Lead *Oct 2024 – Dec 2024*

- Designed and implemented a simple digital counter in VHDL that counts from 0 to 9 and resets.
- Led simulation and synthesis of the design using Intel Quartus for FPGA deployment.
- Applied modular design and state machine logic to ensure performance and maintainability.

Traffic Light System | 555 Timer, 4017 Decade Counter
Team Member *Mar 2024 – May 2024*

- Designed a traffic light circuit using a 555 timer for pulse generation and a 4017 counter for LED sequencing.
- Simulated traffic signal transitions through sequential logic.
- Implemented and tested the circuit on a breadboard using discrete components for real-time demonstration.

AWARDS & CERTIFICATES

- ***The Fundamentals of Digital Marketing – Google***
- ***Received Academic Excellence Award – June 2022***